



iHUB DivyaSampark



EXECUTIVE POST GRADUATE CERTIFICATION IN DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Become skilled in the field of data science and artificial intelligence by enrolling for this course



10 Million
Learners



1:1 Personalized
Mentorship



55% Average
Salary Hike

Executive Post Graduate Certification in Data Science and AI

Master machine learning and artificial intelligence skills with this advanced data science and artificial intelligence course from iHUB IIT Roorkee. Learn from IIT faculty and industry experts with 1:1 mentorship in this intensive online bootcamp. The top 2 performers from each batch may get a fellowship worth Rs. 80,000, plus the opportunity to showcase their startup ideas and secure incubation support of up to Rs. 50 Lakhs for their startup from iHUB DivyaSampark, IIT Roorkee.



Hottest Job of 21st Century



2.7 Million Job Postings

There is a global estimate of 2.7 million job postings for analytics and data science roles by 2022



Skill Development

Data Scientists are equipped with various relevant skills which fetch them lucrative job offers



Growing Analytics Industry

28.8% CAGR in Indian Analytics Industry



Future-oriented Career

Data Science and Artificial Intelligence is a budding field; a head start will prove to be beneficial



Popular Degree

50% of the Data Scientists have a Master's Degree



High Demand

By 2022, India & US will face a demand supply gap of 200,000 Data Science professionals

Our Credentials



10 Million+

Aspiring Students



1,000+

Industry-expert Instructors



400+

Hiring Partners



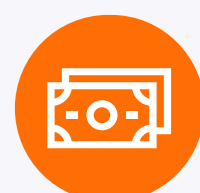
500+

Corporates Upskilled



55%

Average Salary Hike



155+

Countries' Learners

About Program

Become an expert in Data Science and Artificial Intelligence with iHUB IIT Roorkee. Build a strong foundation by learning in-demand skills, solving real-world challenges, and working on real-world projects. A flexible learning environment and 24/7 support make this course perfect for beginners and professionals alike.



Learning Format
Online Bootcamp



11 Months
Live Classes



Campus Immersion
iHUB, IIT Roorkee



Executive PG
Certification

Key Highlights

- ✓ 620 Hrs of Applied Learning
- ✓ 218 Hrs of Self-paced Learning
- ✓ 50+ Industry Projects & Case Studies
- ✓ 24*7 Support
- ✓ 1:1 Mock Interview
- ✓ iHUB DivyaSampark, IIT Roorkee Certification
- ✓ 2 Days Campus Immersion at iHUB, IIT Roorkee
- ✓ 3 Guaranteed Job Interviews upon movement to Placement Pool
- ✓ 90+ Live Sessions Across 11 months
- ✓ Learn from IIT Faculty & Industry Practitioners
- ✓ One-on-One with Industry Mentors
- ✓ Dedicated Learning Management Team
- ✓ No-Cost EMI Option
- ✓ Designed for Working Professionals and Freshers
- ✓ Up to Rs. 50 Lakhs startup Incubation Support*

Program Pedagogy



Instructor-led Training
Get trained by top industry experts



Dedicated Learning Management Team
To help you with your learning needs



Self-paced videos
Learn at your own pace with world-class content



Projects and Exercises
Get real-world experience through projects



Hackathons
Get a sense of how real projects are built



Peer Networking and Group Learning
Improve your professional network and learn from peers



Gamified Learning
Get involved in group activities to solve real-world problems



1:1 Personalized Learning
Hands-on exercises, project work, quizzes, and capstone projects

Who Can **Apply for the Course?**

- ✓ Individuals with a bachelor's degree and a strong interest in learning AI and data science
- ✓ IT professionals looking to make a career transition as data scientists and artificial intelligence engineers
- ✓ Software Developer, Project managers, Non-Technical Professionals & Entry-level professionals looking to build their careers in artificial intelligence and data science
- ✓ Undergraduate freshers with an interest in Data Science & AI

Application **Process**

The application process consists of three simple steps. Candidates have to submit their application. An offer of admission will be made to the selected candidates, and their application will be accepted upon the payment of the admission fee.

1

SUBMIT APPLICATION

Tell us a bit about yourself and why you want to join



2

ADMISSION TEST & APPLICATION REVIEW

Clear the admission test and have a personal interview with our interview panel



3

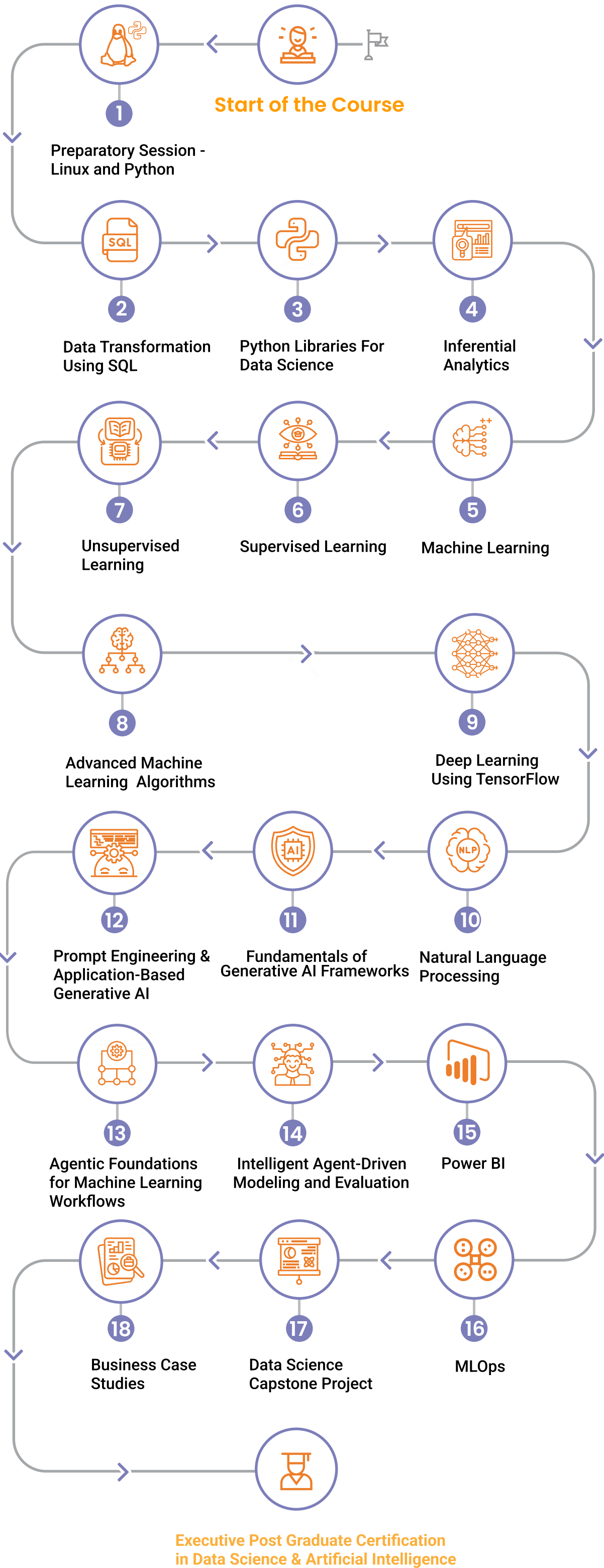
ADMISSION LETTER

Shortlisted candidates would be offered the admission letter



Live Courses

- Live Courses
- Self-paced
- Industry Expert
- Academy Faculty



Self-paced Course

- Data Science At Scale with Spark
- Excel Analytics

Module 1

Preparatory Sessions – Python & Linux

Python

- Introduction to Python and IDEs
- Python Basics
- Object Oriented Programming
- Hands-on Sessions And Assignments for Practice

Linux

- Introduction to Linux
- Linux Basics
- Hands-on Sessions And Assignments for Practice

Module 2

Data Transformation Using SQL

- SQL Basics
- Advanced SQL
- Deep Dive into User Defined Functions
- SQL Optimization and Performance

Module 3

Python Libraries For Data Science

- Data Handling with NumPy
- Data Manipulation Using Pandas
- Data Preprocessing
- Data Visualization

Module 4

Inferential Analytics

- Statistics and Descriptive Analytics using MS Excel
- Python for Descriptive, Diagnostic, and Inferential Statistics
- Prescriptive Analytics

Module 5

Machine Learning

- Introduction to Machine learning
- Regression
- Classification
- Clustering

Module 6

Supervised Learning

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors
- Time Series Forecasting
- Performance Metrics
- Classification reports
- Confusion matrix

Module 7

Unsupervised Learning

- K-means
- Dimensionality reduction
- Linear Discriminant Analysis
- Principal Component Analysis

Module 8

Advanced Machine Learning Algorithms

- Bagging And Boosting Algorithms
- Other Machine Learning Algorithms
- Predictive Analytics And Machine Learning
- Cognitive Science and Analytics

Module 9

Data Science At Scale with spark

- Introduction to Big Data And Spark
- RDDs
- Advanced Concepts & Spark-Hive

Module 10

Deep Learning Using TensorFlow

- Artificial Intelligence Basics
- Neural Networks
- Deep Learning

Module 11

Natural Language Processing

- Text Mining, Cleaning, and Pre-processing
- Text classification, NLTK, sentiment analysis, etc
- Sentence Structure, Sequence Tagging, Sequence Tasks, and Language Modeling
- AI Chatbots and Recommendations Engine

Module 12

Fundamentals of Generative AI Frameworks

- LSTM
- Transformers
- BERT
- GPT
- LLM
- VAEs
- Langchain

Module 13

Prompt Engineering & Application-Based Generative AI

- Prompt Engineering
- Image-Based Applications of Generative AI
- Text-Based Applications of Generative AI
- Audio-Based Applications of Generative AI

Module 14

Agentic Foundations for Machine Learning Workflows

- Understand the meaning of Agentic AI, and why it matters for ML workflows
- Know the role of reasoning, memory, and tool use in building AI agents
- Understand Reasoning + Action (ReAct) + Toolformer + CoT prompting
- Learn to build Agno or LangChain Agents with Pandas for data preprocessing and Scikit-learn for model selection and training
- Hands-on
 -

Module 15

Intelligent Agent-Driven Modeling and Evaluation

- Learn to add decision logic for model suggestion & hyperparameter selection
- Decide how to use memory and intermediate outputs for model building

- Get to know how to autonomously evaluate multiple models and summarise results.
- Get introduced to API-based tools via MCP (Model Context Protocol)
- Hands-on: Learn to build agents for:

Module 16

Power BI

- Power BI Basics
- DAX
- Data Visualization with Analytics
- Hands-on Exercise

Module 17

MLOps

- Introduction to MLOps
- Deploying Machine Learning Models

Module 18

Data Science Capstone Project

Module 19

Business Case Studies

- Recommendation Engine
- Rating Predictions
- Census
- Housing
- Object Detection
- Stock Market Analysis
- Banking Problem
- AI Chatbot

Module 20

Excel Analytics

- Excel Fundamentals
- Excel For Data Analytics
- Data Visualization with Excel
- Excel Power Tools
- Classification Problems using Excel
- Information Measure in Excel
- Regression Problems Using Excel



Skills to Master

- | | | | |
|-----------------|--------------------|-------------------------|----------------------|
| • Python | • Data Science | • Data Analysis | • AI |
| • GIT | • MLOps | • Data Wrangling | • SQL |
| • Story Telling | • Machine Learning | • Prediction algorithms | • NLP |
| • PySpark | • Model | • Data visualization | • Azure Data Factory |

Tools to Master



Projects cover the following industries:



Retail



Social Media



Supply Chain



Entrepreneurship



E-commerce



Banking



Healthcare



Insurance

Real Estate Analytics

Beginner

Real Estate Analytics will involve supervised learning with an ensemble of various regression algorithms where we will optimize the predictions based on the error rate.

Solar Power Efficiency

Beginner

The project will encompass three target variables that we will predict using the supervised machine learning algorithms for regression problems and minimize the error by tuning the hyperparameters.

Recommendation Engine

Beginner

In the Recommendation Engine project, we will use singular value decomposition to draw out relevant recommendations for music and movie selections based on the historical data points.

Stock Price Prediction

Intermediate

Forecasting a stock price based on time series analysis and forecasting using ARIMA, Seasonal ARIMA and other deep learning techniques to forecast stock prices based on historical data points.

Weather Forecasting

Intermediate

Ensemble of supervised machine learning algorithms to predict and forecast the weather based on historical data points, & optimization of the model to reduce the error in predictions using hyperparameter tuning and other data modeling techniques.

Image Classification

Intermediate

The project involves processing image data that is fed to the convolutional neural networks which then efficiently predicts the species of the input that is provided during the predictions.

Gesture Recognition

Advance

In this computer vision project, we will extract frames in real time and feed it to an already trained neural network that will correctly classify the gesture that is most relevant to the specified frame in real time.

American Sign Language Recognition

Advance

In this project we will use computer vision and deep learning to feed real time frames that will correctly predict the ASL references to the signs made in the video and generate the audio for the same as well.

Sentiment Analysis

Advance

Classic NLP problem encompasses processing textual data and then feeding the information to the deep learning model which then processes and predicts the sentiment associated with the text in both supervised and unsupervised learning settings.

Housing Price Prediction

Advance

In this project on housing price prediction, get practical exposure to how to work with a house price dataset and predict the sale price for each house with 79 explanatory variables describing every aspect of the houses.



Dr. Sudeb Das Gupta

Professor at IIT Roorkee

Dr. Sudeb Dasgupta is currently working as Professor at IIT Roorkee. He has completed his PhD from Banaras Hindu University in Electronics Engineering and has more than 20 years of teaching experience.



Dr. Balasubramanian R

Professor at IIT Roorkee

Dr. Balasubramanian R, is currently working as Professor at IIT Roorkee. He has completed his PhD from IIT Madras in Mathematics and Computer Science and has more than 20 years of teaching experience. His areas of interest include computer vision, machine learning, pattern recognition, deep learning, etc.



Dr. Narendra S. Chaudhari

Professor at IIT Indore

Dr. Narendra Chaudhari, Professor of Computer Science and Engineering at IIT Indore, Dr. Narendra Chaudhari, a Professor at IIT Indore, specializes in network security, mobile computing, and machine learning. He holds a Ph.D. from IIT Bombay and has previously served as the Director of VNIT Nagpur and Vice Chancellor of Uttarakhand Technical University



Dr. Sushanta Karmakar

Associate Professor at IIT Guwahati

Dr. Sushanta Karmakar is currently working as Associate Professor, at IIT Guwahati. He has completed his PhD from IIT Kharagpur from the Computer Science and Engineering department and has over 15 years of teaching experience.



Chandresh Kumar Maurya

Professor at IIT Indore

Chandresh Maurya, a dynamic professor at IIT Indore with over 12 years of experience, excels in driving innovation and growth through his expertise in multimodal data processing, AI, ML, and DL. He has a proven track record of leading cross-functional teams and managing complex projects to deliver impactful solutions. His strategic vision and ability to tackle challenges have consistently enhanced operational efficiency and advanced research in Speech and NLP.



Madhu Babu Cherukuri

Sr. Data Scientist at Intel Corporation

He carries over 15 years of expertise in handling various analytics projects, using several types of ML algorithms in multiple business domains. He also comes with extensive knowledge in SAS, SQL, R, Python and is proficient in Big Data Environment.



Phaneendra Akula

Senior Manager, Data Science at Sunrise Systems

Mr. Akula is presently working at Sunrise Biztech Systems Private Limited. He has 9+ years of experience in the field of Data Science and has worked in many prestigious companies like Genpact, TCS, and Tesco. He is also a research scholar and a speaker.



Nishkam Shivam

Senior Data Scientist at Bristlecone

A Senior Data Scientist in the field of Analytics and AI, Mr. Nishkam Shivam is presently working at Bristlecone. He has 9+ years of experience in this particular field and has also worked in renowned companies like Accenture, Walmart, Emeritus, and many more.



Anurag Singh

Principal AI Architect at Schneider Electric Hub

Mr. Singh works at Schneider Electric Hub as a Principal AI Architect. He started as a Software Engineer in 2007 and has been working as Data Scientist since 2018. He specializes in Big Data, Data Science, Machine Learning, and AI System Architecture.



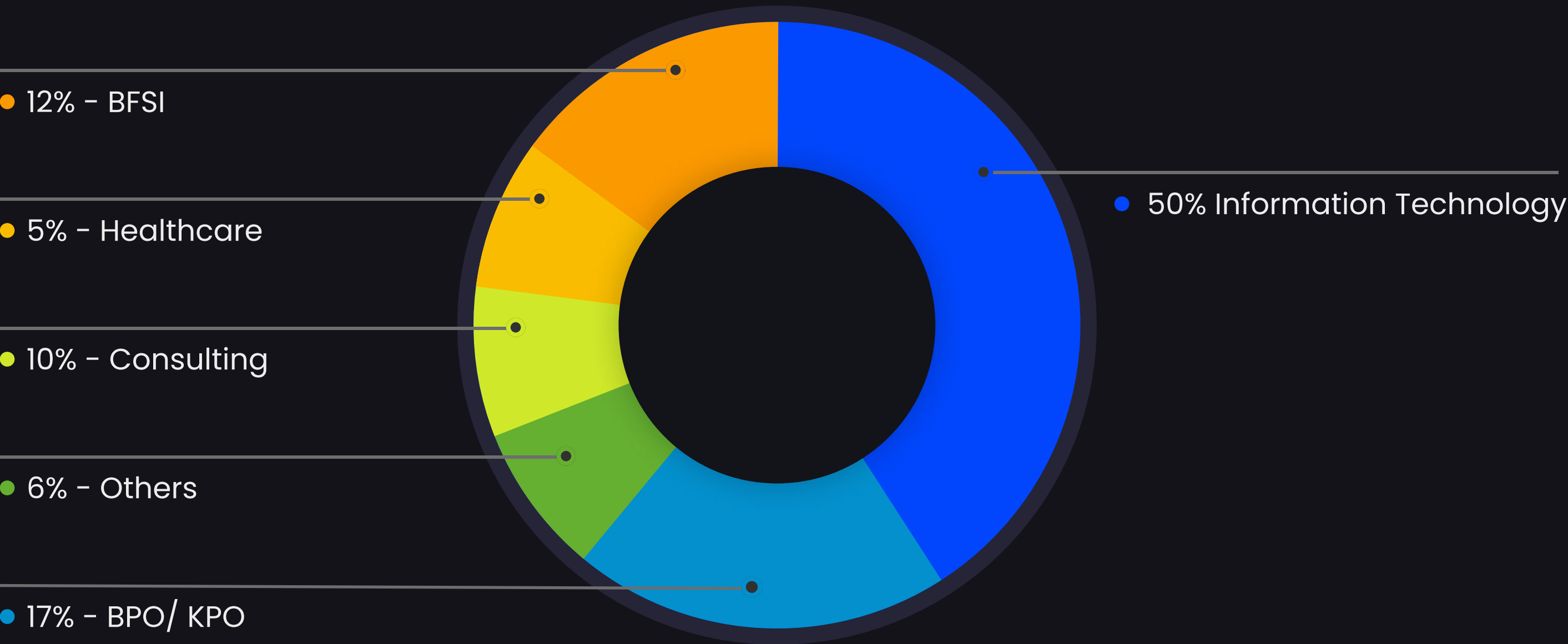
Vishwath Parthasarathy

Data Scientist-III at InMobi

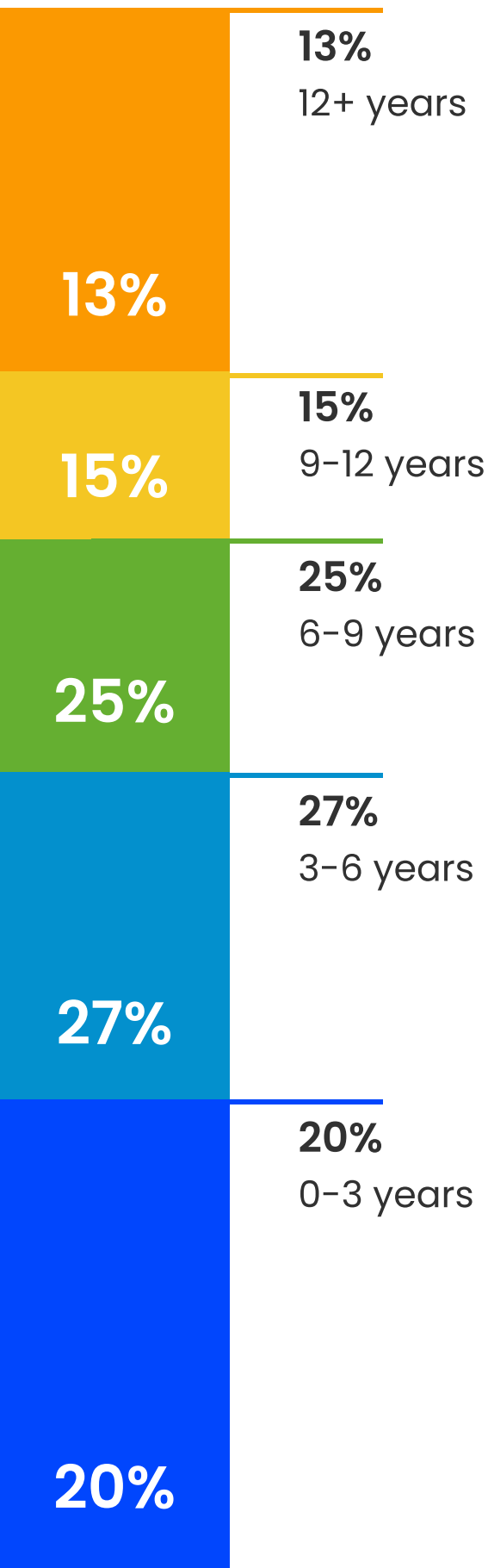
Mr. Parthasarathy is currently working as a Data Scientist-III at InMobi. He has also worked at Amazon and PipeCandy. His expertise extends to Machine Learning, Deep Learning, MLOps, and Algorithmic Trading. He has been working since 2017 as a Data Scientist.

Meet the Batch

Industries Our Learners Come From



Work Experience



Some of their current employers include

Intellipaat Career Services



500+
Webinars



600+
Job Shares



400+
Hiring Partners



55%
Avg. Salary Hike*

What Makes Us **Tick**



Career-oriented Sessions

Attend 10+ career-oriented sessions by industry mentors and prepare your career trajectory



Profile Building

Craft a resume highlighting Data Science & AI and a LinkedIn profile to make an impression on top employers



Mock Interview Preparation

Prepare yourself with mock interviews including the most asked questions by top employers



1:1 Mentoring Sessions

Get 1:1 guidance at every step in your career transition to Data Science and AI



Placement Assistance

Placement opportunities are provided once the learner is moved to the placement pool upon clearing Placement Readiness Test (PRT)**



Dedicated Job Portal Access

Get exclusive access to 200 job postings per month on Intellipaat's job portal



Job Fairs

Job fairs are conducted regularly to introduce learners to major organizations



Hackathons

Work in teams and get exclusive access to hackathons

Learner **Reviews**



Anoop Prasad

Assistant Professor at EWIT

The training and support team are highly cooperative. The best thing about it is the prompt support. The trainers are well-versed with the concepts and great content.



Manas Ranjan

Data Analyst at Amazon

I am happy to have enrolled in this data science program. The syllabus is organized and the course is well designed. Best features are the 24*7 support and trainers who are domain experts.



Shreyash Limbhetwala

Technical Delivery Lead

Intellipaat offers a well curated rich learning content which I liked the most. The set of PPTs, PDFs, and other course material were of highest quality, due to which my learning was great too.

Glimpse of our Successful Transitions



4.6



4.5



4.38

Career Transitions



Gaurav Saboo
Senior Technical Associate

From
Technical Analyst

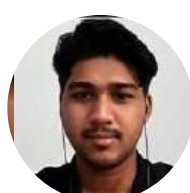
To
Senior Technical Associate



Melwin Rodrigues
Data Scientist

From
Customer Service Agent

To
Data Scientist



D Rajesh Sahoo
Machine Learning Engineer

From
B.Tech – Civil Engineering

To
Machine Learning Engineer



Ankit Kumar
Data Scientist

From
B.com Graduate

To
Data Scientist



Mahesh Chowdary
Data Scientist

From
Aircraft Maintenance Engineer

To
Data Scientist



Jeanette Masso
Big Data Developer

From
Computer Technical Specialist

To
Big Data Developer



Prosenjeet Saha
Data Scientist

From
Manager Analytics & Data Science

To
Data Scientist





About IntelliPaat

IntelliPaat is one of the leading online training providers, with more than 10 million learners in over 155 countries. We are on a mission to democratize education because we believe that everyone has the right to quality education.

We create courses in collaboration with top universities and MNCs focused on employability, including iHUB IIT Roorkee, EICT IIT Guwahati, IITM Pravartak, IIM Ranchi, IIT Jammu, VNIT Nagpur, IU Germany, University of Essex, University of Liverpool, Microsoft, and others.

Our courses come with 24/7 support and are delivered by Subject Matter Experts (SMEs). These courses help learners increase their employability, and our outstanding career services help them jump-start their careers in their dream jobs.



About iHUB DivyaSampark, IIT Roorkee

iHUB DivyaSampark, IIT Roorkee aims to enable an innovative ecosystem in new-age technologies like AI, ML, Drones, Robots, and data analytics (often called CPS technologies). It aims to do this by promoting and enhancing core competencies, capacity building, and manpower training to provide solutions for national strategic sectors and become a significant contributor to Digital India.

Upon the completion of this program, you will:

- Certification from iHUB DivyaSampark, IIT Roorkee



Collaboration with Microsoft

When it comes to new-age technologies and tools, Microsoft is the biggest names and leading innovators. The course is led by experts who will walk you through all of the necessary skills in the domain and get you started on projects relevant to various organisations.



10 Million Learners & 500+ Corporates across 155+ countries
upskilling on Intellipaath Platform



Contact Us

INDIA

6th Floor, Primeco Towers, Arekere Gate Junction, Bannerghatta
Main Road, Bengaluru, Karnataka 560076, India
Phone No: +91-7022374614

USA

INTELLIPAATH INC
300 DELAWARE AVE STE 210-207
WILMINGTON, DE 19801

sales@intellipaath.com

www.intellipaath.com

